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December 21, 2020

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

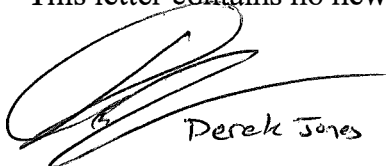
**SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387/2020-003-00
UNIT 1 LICENSE NO. NPF-14
PLA-7913**

Docket No. 50-387

Attached is Licensee Event Report (LER) 50-387/2020-003-00. The LER reports an event involving entry into the mode of applicability without meeting the required Limiting Conditions for Operation. The condition is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by the plant's Technical Specifications.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.


Derek Jones for K. Cimorelli
K. Cimorelli

Attachment: LER 50-387/2020-003-00

Copy: NRC Region I
Mr. C. Highley, NRC Senior Resident Inspector
Ms. S. Goetz, NRC Project Manager
Mr. M. Shields, PA DEP/BRP

**LICENSEE EVENT REPORT (LER)**

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form <https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Susquehanna Steam Electric Station

2. Docket Number

05000387

3. Page

1 of 4

4. Title

Condition Prohibited by Technical Specifications due to Inoperable Turbine Stop Valve and Turbine Control Valve Instrumentation

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
05	09	2020	2020	- 003 -	00	12	21	2020	Facility Name	Docket Number
										05000
									Facility Name	Docket Number
										05000

9. Operating Mode

1

10. Power Level

24

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ Other (Specify here, in Abstract, or in NRC 366A).**12. Licensee Contact for this LER****Licensee Contact**

Shane Jurek, Senior Engineer

Phone Number (Include Area Code)

(570) 542-1695

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	SB	PSF	B130	Y	X	SB	PSF	F155	Y

14. Supplemental Report Expected☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)**15. Expected Submission Date**

Month	Day	Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On May 9, 2020, operators at the Susquehanna Steam Electric Station (SSES), Unit 1, were raising power following a maintenance outage. SSES Technical Specifications (TS) require that certain Turbine Stop Valve (TSV) and Turbine Control Valve (TCV) instrumentation be operable at greater than or equal to 26 percent Rated Thermal Power (RTP). However, at approximately 15:23, Unit 1 reached 26 percent RTP and the TSV and TCV instrumentation was found to be inoperable. Operators declared the TSV and TCV instrumentation inoperable, and entered Conditions A and B of Limiting Condition for Operation (LCO) 3.3.1.1. Control rods were inserted and Unit 1 was brought below 26 percent RTP at approximately 15:37, thereby exiting Conditions A and B of LCO 3.3.1.1. Based on the investigation and identified cause of this event, the inoperable condition likely existed prior to entering the Mode of Applicability. LCO 3.0.4 prohibits reaching 26 percent RTP when the LCO 3.3.1.1 is not met. This event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS.

The cause of the event was determined to be two socket weld failures caused by high cyclic fatigue. Additionally, one of the socket welds was found to be undersized for its application. This condition was corrected by repairing the welds and upgrading the welds to the proper size. There were no actual safety consequences associated with the described condition.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Susquehanna Steam Electric Station, Unit 1	05000-387	2020	- 003 -	00

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 24 percent Rated Thermal Power (RTP)
Unit 2 – Mode 1, approximately 98 percent RTP

After the event, two steam leaks were discovered on instrument piping [EIS System/Component Code: SB/PSF] leading from Turbine Control Valves (TCVs) 2 and 3 [TA/FCV] to turbine pressure switches [IT/PS] which feed into the Reactor Protection System [System Code JC]. The steam leaks rendered Unit 1 Turbine Stop Valve (TSV) [TA/SHV] and TCV instrumentation inoperable prior to the start of the event.

EVENT DESCRIPTION

On May 9, 2020, operators at the Susquehanna Steam Electric Station (SSES), Unit 1, were raising power following a maintenance outage. At approximately 15:19, with reactor power at 24 percent RTP, operators withdrew control rods to raise power to approximately 28 percent RTP. SSES Technical Specification (TS) Limiting Condition for Operation (LCO) 3.3.1.1, "Reactor Protection System (RPS) Instrumentation," requires that TSV and TCV instrumentation be operable (i.e., not bypassed) at greater than or equal to 26 percent RTP. However, at approximately 15:23, Unit 1 reached 26 percent RTP and the TSV and TCV instrumentation was still bypassed. Operators immediately stopped withdrawing control rods, declared the TSV and TCV instrumentation inoperable, and entered Conditions A and B of LCO 3.3.1.1. Unit 1 was brought below 26 percent RTP at approximately 15:37, thereby exiting Conditions A and B of LCO 3.3.1.1 after approximately 14 minutes.

Subsequent to reducing power below 26 percent RTP, Susquehanna discovered two steam leaks on instrument piping lines SPDCD117-2 (specifically from Field Weld-10 (FW-10)) and JD-5-3-1A (specifically from FW-60). These leaks resulted in the instrument lines being unable to provide adequate steam pressure to turbine instrument sensing lines, and were determined to have rendered the TSV and TCV instrumentation inoperable during startup. Susquehanna repaired the leaks by installing new socket welds on May 11, 2020, and resumed power ascension.

The Completion Times associated with LCO 3.3.1.1, Required Actions A.1 and B.1 (12 hours and 6 hours, respectively), were not exceeded. However, SSES TS LCO 3.0.4 prohibits entering a mode of applicability (i.e., reaching 26 percent RTP) without meeting all required LCOs. Based on the investigation and determined cause for this condition, the instrumentation piping likely failed, rendering the TCV and TSV instrumentation inoperable, prior to entry into the Mode of Applicability for LCO 3.3.1.1. As such, this event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS.

CAUSE OF EVENT

The cause of the event was determined to be FW-10 and FW-60 socket weld failure due to high cyclic fatigue. Additionally, the weld on FW-10 was determined to be undersized.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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Susquehanna Steam Electric Station, Unit 1	05000-387	2020	- 003 -	00

NARRATIVE**ANALYSIS/SAFETY SIGNIFICANCE**

The inoperable TSV and TCV instrumentation was limited to a single RPS subsystem; the other subsystem was capable of initiating a valid reactor scram signal. As such, there was no loss of safety function (i.e., a safety system functional failure did not occur). This event did not result in any actual consequences to the health and safety of the public.

CORRECTIVE ACTIONS

Corrective actions for this event included repairing FW-10 and FW-60 by removing the leaking connections and installing new welds.

COMPONENT FAILURE INFORMATION**FW-10**

Component Identification – SPD CD117-2

Component Name – Pressure Lines from Main Steam Line TCV-2 and TCV-3 to Turbine Pressure Instruments

Component Description – 1" Schedule XXS Instrument Sensing Line

Component Manufacturer – Bechtel

Socket Weld Location – Downstream end of Valve 101009

FW-60

Component Identification – JD-5-3-1A

Component Name - Pressure Lines from Main Steam Line TCV-2 and TCV-3 to Turbine Pressure Instruments

Component Description – 0.5" Flex Hose

Component Manufacturer – Senior Flexonics

Socket Weld Location – Connection of 0.5" Flex Hose to 0.5" Reducer

PREVIOUS OCCURRENCES

LER 50-388/2015-004-00, "Degraded Condition due to Reactor Coolant Pressure Boundary Leakage Caused by Vibration and Stiff Pipe Connection," dated June 10, 2015.

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Susquehanna Steam Electric Station,
Unit 1

2. DOCKET NUMBER

05000-387

3. LER NUMBER

YEAR

SEQUENTIAL
NUMBER

REV
NO.

2020

- 003 -

00

NARRATIVE

LER 50-387/2014-011-00, "Degraded Condition Due to Reactor Coolant Pressure Boundary Leakage Caused by an Inadequate Weld," dated February 11, 2015.

LER 50-387/2012-007-01, "Unplanned Shutdown due to Unidentified Drywell Leakage," dated November 20, 2012.